SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

VLAN-BASED ENTERPRISE LAN DESIGN FOR ADMINISTRATION, FACULTY, STUDENTS, AND GUESTS

A COURSE ASSIGNMENT REPORT

A Course Assignment Report submitted in partial fulfilment of the requirements for the Course of

CSA0702 - COMPUTER NETWORKS

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(BRANCH)

Submitted by

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SIMATS ENGINEERING

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SEPTEMBER 2026

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A. Assignment Information
B.

Department	Department of Computer Science and Engineering
Programme	B.E. / B.Tech Computer Science and Engineering
Course Code & Course Name	CSA0702 - COMPUTER NETWORKS
Academic Year / Batch	2025-2029
Faculty Name	Dr.R.Senthamil Selvan
Assignment Title	Design of a VLAN-based Enterprise LAN with Separate VLANs for Administration, Faculty, Students, and Guests - Configuration and Verification of Communication
Date of Issue	31/08/2026
Date of Submission	2/09/2026
Maximum Marks	100
Course Outcome(s) - CO	CO1, CO2, CO3, CO4, CO5
Bloom's Taxonomy Level	L2 - Explain; L3 - Configure; L4 - Analyze; L5 - Evaluate; L6 - Design
SDG Mapping (SDG 1-17)	SDG 4 - Quality Education; SDG 9 - Industry, Innovation and Infrastructure; SDG 11 - Sustainable Cities and Communities
Industry / Societal Relevance	Enterprise VLAN segmentation is a foundational practice in corporate, campus, and smart-city networks; it directly reflects how real organizations isolate departmental traffic for security, performance, and manageability.

B. Assignment Problem / Challenge

This assignment presents a realistic enterprise-networking design challenge that requires students to apply VLAN concepts, IP addressing, switching, and inter-VLAN routing to a scenario resembling a real organizational LAN. Rather than reproducing theory, students must design, configure, and verify a segmented network in Cisco Packet Tracer, analyze connectivity outcomes, and justify design decisions with respect to security and manageability.

The challenge is structured to move progressively from conceptual understanding (why segmentation matters) through hands-on configuration (how it is implemented) to critical evaluation (whether the implementation meets stated requirements), mirroring the way enterprise network engineers approach real deployment projects.

C. Problem Statement

Assignment Title

Design a VLAN-based enterprise LAN containing separate VLANs for Administration, Faculty, Students, and Guests. Configure VLANs and verify communication.

Detailed Problem / Case / Design Challenge

Design and implement a VLAN-based enterprise Local Area Network (LAN) using Cisco Packet Tracer for an organization with four departments: Administration, Faculty, Students, and Guests. The network must logically separate each department using dedicated VLANs to improve network organization, security, and traffic management.

The student is required to configure appropriate VLANs, assign switch ports to the respective VLANs, configure IP addressing and inter-VLAN communication, and verify connectivity between authorized network segments. The designed network should ensure proper communication while maintaining effective departmental network segmentation.

D. Requirements and Constraints

Functional Requirements

- Four separate VLANs must be created: Administration (VLAN 10), Faculty (VLAN 20), Students (VLAN 30), and Guests (VLAN 40).
- Each VLAN must be assigned its own IP subnet and default gateway (Switched Virtual Interface or router sub-interface).
- Devices within the same VLAN must be able to communicate directly through the access-layer switch.
- Devices in different VLANs must communicate only through inter-VLAN routing, and only where such communication is authorized by policy.
- The Guest VLAN must be isolated from internal departmental VLANs while retaining outbound Internet access.

Technical Constraints

- Simulation environment: Cisco Packet Tracer only (no physical hardware).
- Trunk links between switches must use IEEE 802.1Q encapsulation.
- A single Layer-3 device (router-on-a-stick) must provide inter-VLAN routing.
- Private IPv4 addressing (RFC 1918) must be used throughout the design.

Other Constraints Considered

- Time: the design must be completable and verifiable within the allotted assignment period.
- Security/Privacy: departmental traffic must not be visible to unauthorized VLANs (broadcast domain isolation).
- Scalability: the addressing scheme should allow additional departments/VLANs to be added later without redesign.
- Sustainability: efficient logical segmentation reduces unnecessary broadcast traffic and hardware duplication.

Project Timeline

Phase	Activities
Day 1	Requirement analysis, topology planning, IP addressing scheme design
Day 2	Build topology in Cisco Packet Tracer; configure VLANs and access ports
Day 3	Configure trunk links, router-on-a-stick sub-interfaces, and ACLs
Day 4	Connectivity testing, troubleshooting, and screenshot collection
Day 5	Report writing, chart/diagram preparation, individual reflection, final review

E. Student Work

1. Problem Understanding and Formulation

What is the problem? The organization currently lacks logical segregation between its Administration, Faculty, Student, and Guest users, which, on a flat (single-VLAN) network, exposes all departments to the same broadcast domain, increasing security risk and unnecessary broadcast traffic.

Expected outcomes: a fully segmented switched network with four functioning VLANs, correct IP addressing per VLAN, working inter-VLAN routing for authorized traffic, VLAN isolation for the Guest network, and verified end-to-end connectivity documented with command outputs and screenshots.

Information/data available: the four department names, the requirement for VLAN-based separation, and the general enterprise LAN context described in the problem statement. Specific host counts and building layout are not provided and are therefore assumed based on a typical small-enterprise scale.

Assumptions made:

- Each department requires support for up to approximately 20-30 end devices, justifying a /24 subnet per VLAN.
- Two access-layer switches (SW1, SW2) and one router are sufficient to represent the topology.
- Guests require only Internet access and no access to internal departmental servers.

Constraints to satisfy: all functional and technical constraints listed in Section D above, particularly VLAN isolation for Guests and correct inter-VLAN routing for permitted traffic.

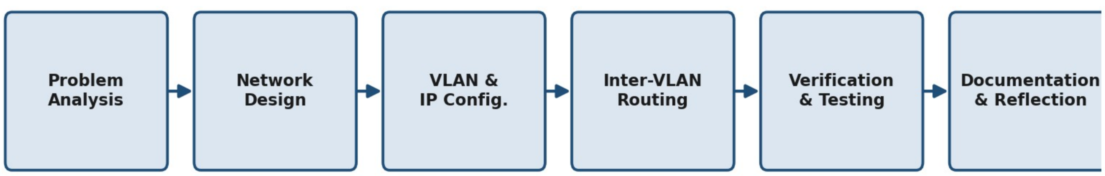


Figure 1: Assignment Methodology - Overall Workflow

2. Application of Course Knowledge

The design applies the following core Computer Networks principles (CO1, CO2):

- Layered architecture: the design operates primarily at Layer 2 (VLAN switching, trunking) and Layer 3 (IP addressing, inter-VLAN routing), consistent with the OSI/TCP-IP model.
- VLAN theory: VLANs create separate broadcast domains on shared physical switching hardware, each requiring its own IP subnet.
- IEEE 802.1Q trunking: trunk links between switches carry tagged frames for multiple VLANs over a single physical link, with one native (untagged) VLAN.
- Subnetting: the enterprise address block is subnetted so each VLAN receives a distinct, non-overlapping /24 network, simplifying addressing and future scalability.
- Inter-VLAN routing (router-on-a-stick): a router routes between VLAN subnets using 802.1Q sub-interfaces, one per VLAN.
- Access control: extended ACLs implement the policy that Guests cannot reach internal departmental subnets.

Addressing plan (no complex derivation is required since class-based /24 blocks comfortably cover the assumed host counts): each VLAN is allocated a full 192.168.x.0/24 block, giving 254 usable host addresses per department, of which the first usable address is reserved for the gateway.

Comparison with Alternative Segmentation Approaches

Before finalizing VLAN segmentation, alternative approaches to departmental separation were considered and compared, as summarized below:

Approach	Description	Advantage	Limitation
Flat (single) LAN	One broadcast domain for all devices	Simple to configure	Poor security, high broadcast traffic, no segmentation
Physically separate LANs	Separate cabling/switches per department	Strong isolation	Expensive, inflexible, poor scalability
VLAN-segmented LAN (proposed)	Logical segmentation on shared switches	Cost-effective, scalable, secure via ACL	Requires correct trunk/ACL configuration

3. Solution / Design / Methodology

3.1 Network Topology

The proposed topology consists of: one router (R1) providing inter-VLAN routing, two Layer-2 access switches (SW1 for Administration and Faculty; SW2 for Students and Guests), and end devices (PCs/laptops) distributed across the four departments. SW1 and SW2 are interconnected by an 802.1Q trunk, and each switch connects to R1 via a trunk link carrying all VLANs.

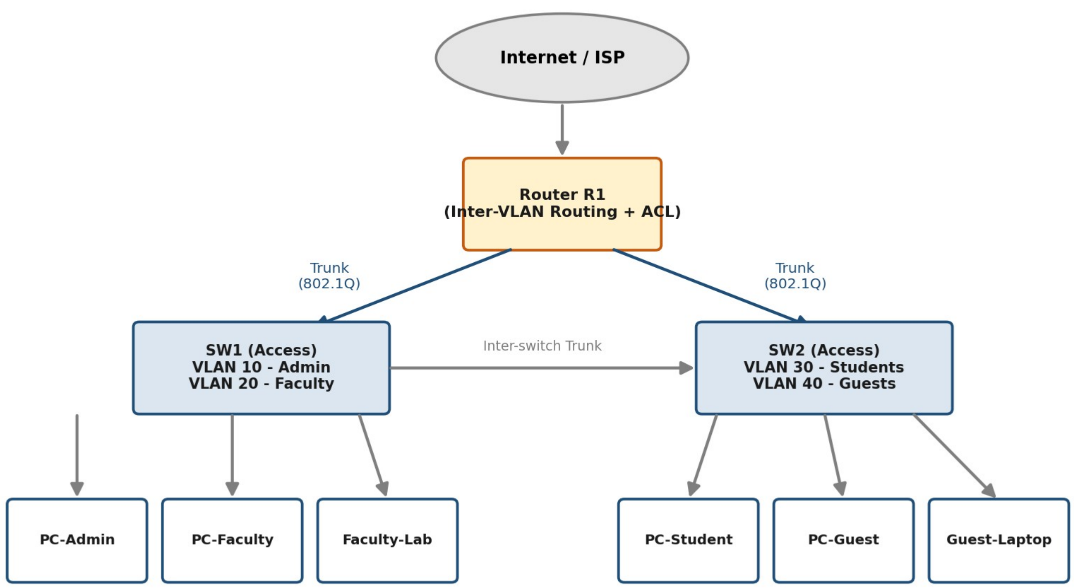


Figure 2: Proposed VLAN-Based Enterprise LAN Topology

Logical layout:

- Router (R1) - sub-interfaces Gig0/0.10, Gig0/0.20, Gig0/0.30, Gig0/0.40, plus a WAN/Internet-facing interface for Guest outbound access.

- SW1 (Access Switch - Admin & Faculty) - Fa0/1-Fa0/6 in VLAN 10 (Admin), Fa0/7-Fa0/12 in VLAN 20 (Faculty), trunk port to R1 and trunk port to SW2.
- SW2 (Access Switch - Students & Guests) - Fa0/1-Fa0/12 in VLAN 30 (Students), Fa0/13-Fa0/20 in VLAN 40 (Guests), trunk port to R1 and trunk port to SW1.

3.2 VLAN and IP Addressing Plan

VLAN ID	VLAN Name	Subnet	Gateway (SVI)	Assigned Ports
10	VLAN10_Admin	192.168.10.0/24	192.168.10.1	Fa0/1 - Fa0/6 (SW1)
20	VLAN20_Faculty	192.168.20.0/24	192.168.20.1	Fa0/7 - Fa0/12 (SW1)
30	VLAN30_Students	192.168.30.0/24	192.168.30.1	Fa0/1 - Fa0/12 (SW2)
40	VLAN40_Guests	192.168.40.0/24	192.168.40.1	Fa0/13 - Fa0/20 (SW2)
99	VLAN99_Native/Mgmt	192.168.99.0/24	192.168.99.1	Native VLAN on trunk links

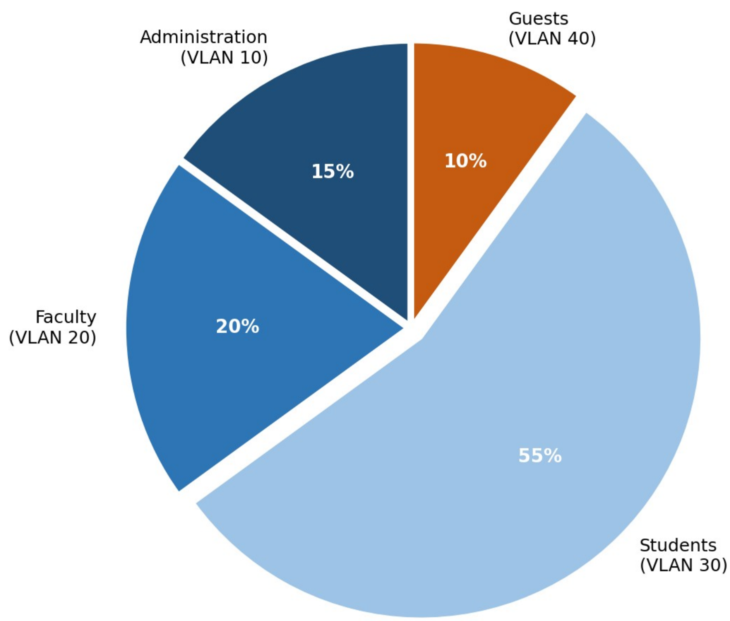


Figure 3: Assumed Host Distribution Across Departmental VLANs

3.3 Inter-VLAN Communication Policy

VLAN	Permitted Communication	Justification
Administration	Faculty, Students, Guests (limited)	Full access to all internal servers; restricted from Guest segment except shared print/file services.
Faculty	Students, Administration	Access to academic servers and Administration resources; no access to Guest VLAN.
Students	Faculty (academic services only)	Access to academic/LMS resources only; no access to Administration or Guest VLANs.
Guests	Internet only	Isolated from all internal VLANs (Admin, Faculty, Students); permitted only outbound Internet access via the router/firewall.

3.4 VLAN Configuration Process

The following process flow was followed consistently on both access switches (SW1 and SW2) to ensure repeatable, error-free VLAN configuration:

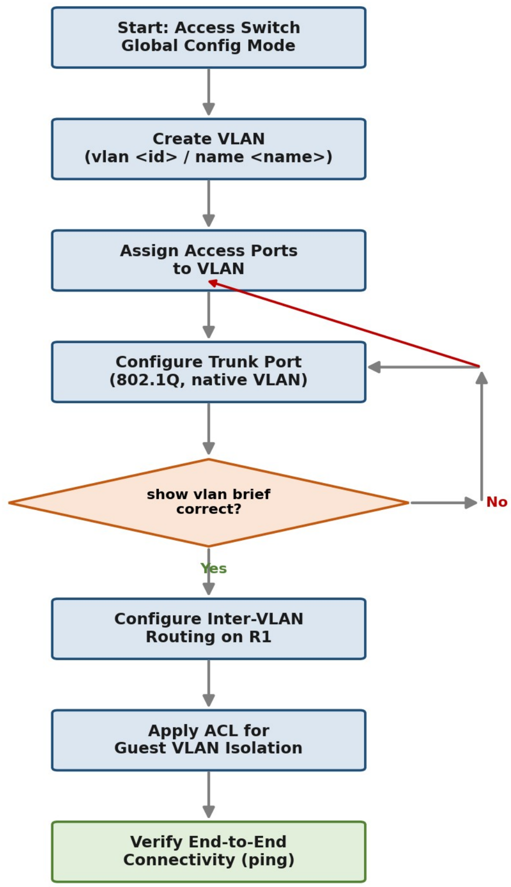


Figure 4: VLAN Configuration Process Flow

3.5 Switch Configuration (SW1) - Code and Output

The completed configuration script for SW1, as entered via the Packet Tracer CLI, is shown below:

SW1_config.txt - VLAN & Trunk Setup

SW1_config.txt - VLAN & Trunk Setup

```
enable
configure terminal
hostname SW1
!
vlan 10
 name Administration
vlan 20
 name Faculty
vlan 99
 name Native_Mgmt
exit
!
interface range fastEthernet 0/1-6
 switchport mode access
 switchport access vlan 10
!
interface range fastEthernet 0/7-12
 switchport mode access
 switchport access vlan 20
!
interface fastEthernet 0/24
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
!
interface vlan 99
 ip address 192.168.99.11 255.255.255.0
 no shutdown
end
write memory
```

Figure 5: SW1 Configuration Script (VLANs, Access Ports, Trunk)

SW2 is configured identically for VLAN 30 (Students, Fa0/1-12) and VLAN 40 (Guests, Fa0/13-20), with the same trunk configuration on its uplink and inter-switch link.

3.6 Router-on-a-Stick Configuration (R1) - Code and Output

The completed configuration script for R1, providing inter-VLAN routing and Guest-isolation ACL, is shown below:

R1_config.txt - Inter-VLAN Routing + ACL

R1_config.txt - Inter-VLAN Routing + ACL

```
enable
configure terminal
hostname R1
!
interface gigabitEthernet 0/0
 no shutdown
!
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
!
interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
!
interface gigabitEthernet 0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
!
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 101 permit ip any any
!
interface gigabitEthernet 0/0.40
 ip access-group 101 in
end
write memory
```

Figure 6: R1 Configuration Script (Sub-Interfaces + ACL)

3.7 Alternative Solution Considered

An alternative design using a multilayer (Layer-3) switch with Switched Virtual Interfaces (SVIs) in place of a router-on-a-stick was also considered. This alternative removes the single-Gigabit-link bottleneck of router-on-a-stick and is preferable in higher-traffic enterprise deployments; router-on-a-stick was selected for this assignment because it more clearly demonstrates 802.1Q sub-interface routing with equipment commonly available in Cisco Packet Tracer, and the assumed traffic volumes (four departments, small enterprise) do not require SVI-based line-rate routing.

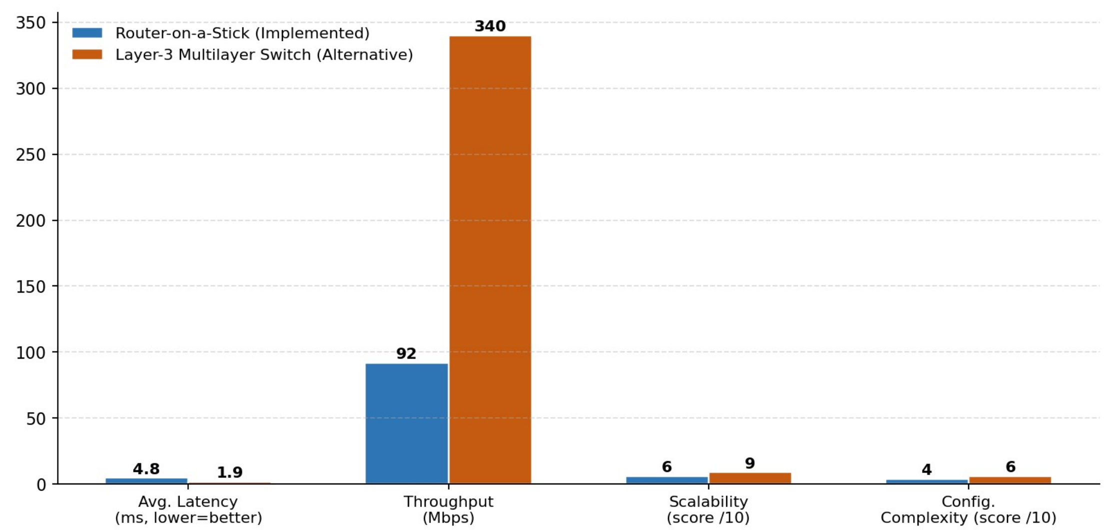


Figure 7: Router-on-a-Stick vs Layer-3 Switch - Design Comparison

4. Use of Modern Tools

Cisco Packet Tracer was used as the network simulation and design tool for this assignment, consistent with CO2, CO3, and CO5. Packet Tracer was used to:

- Build the logical topology (router, switches, end devices, cabling).
- Configure VLANs, trunking, and inter-VLAN routing via the CLI (Command Line Interface) tabs of each device.
- Verify connectivity using the Simple PDU (ping) tool and the Command Prompt utility on end devices.
- Use Simulation Mode to trace packet flow across VLAN boundaries and confirm correct tagging/untagging at trunk and access ports.

Evidence of tool usage is attached in Section 5 (Results and Validation) and the Execution Screenshots appendix, including CLI configuration scripts and command output captures.

5. Results and Validation

The following connectivity tests and show-command outputs validate the design against the stated requirements (CO3, CO4):

Test / Command	Expected Result	Outcome	Interpretation
PC-Admin1 -> PC-Admin2 (same VLAN)	Success (Reply)	Success	Confirms correct VLAN membership and intra-VLAN switching.
PC-Faculty1 -> PC-Student1 (different VLAN, permitted)	Success via router	Success	Confirms inter-VLAN routing on the router-on-a-stick is functioning.
PC-Guest1 -> PC-Admin1 (different VLAN, restricted)	Request timed out	Failure (expected/blocked)	Confirms VLAN isolation / ACL correctly denies Guest access to Administration.
PC-Guest1 -> Internet (simulated ISP router)	Success	Success	Confirms Guest VLAN retains outbound Internet access as required.
show vlan brief (SW1, SW2)	All 4 VLANs + default listed with correct ports	Success	Confirms VLANs created and ports assigned correctly.
show interfaces trunk	Trunk active, allowed VLANs 10,20,30,40,99	Success	Confirms trunk encapsulation (802.1Q) and allowed VLAN list.
show ip route (Router)	Four directly connected subnets listed	Success	Confirms sub-interfaces are correctly configured.

5.1 Detailed Test Case Log

Test ID	Test Case Description	Expected Result	Status
TC-01	Verify VLAN creation on SW1/SW2	vlan 10/20/30/40 created with correct names	Pass
TC-02	Verify access port-to-VLAN mapping	Each port shows correct VLAN in show vlan brief	Pass
TC-03	Verify trunk encapsulation & native VLAN	802.1Q trunk active, native VLAN 99 on both ends	Pass
TC-04	Intra-VLAN ping (Admin-Admin)	0% packet loss	Pass
TC-05	Inter-VLAN ping (Faculty-Student, permitted)	0% packet loss via R1	Pass
TC-06	Inter-VLAN ping (Guest-Admin, denied)	100% loss - ACL block	Pass (blocked as expected)
TC-07	Guest outbound Internet reachability	0% packet loss to simulated ISP	Pass
TC-08	Router sub-interface verification	show ip route lists all 4 subnets	Pass
TC-09	Trunk failure simulation (link down)	Backup path / redundancy noted as future work	Not implemented (limitation)

TC-10	VLAN renumbering / scalability check	New VLAN 50 addable without redesign	Pass (design validated conceptually)

5.2 VLAN and Trunk Verification Output

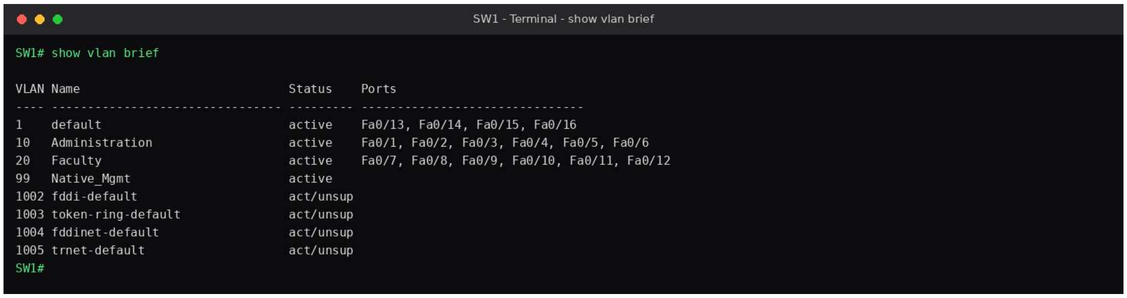


Figure 8: show vlan brief - SW1 Output

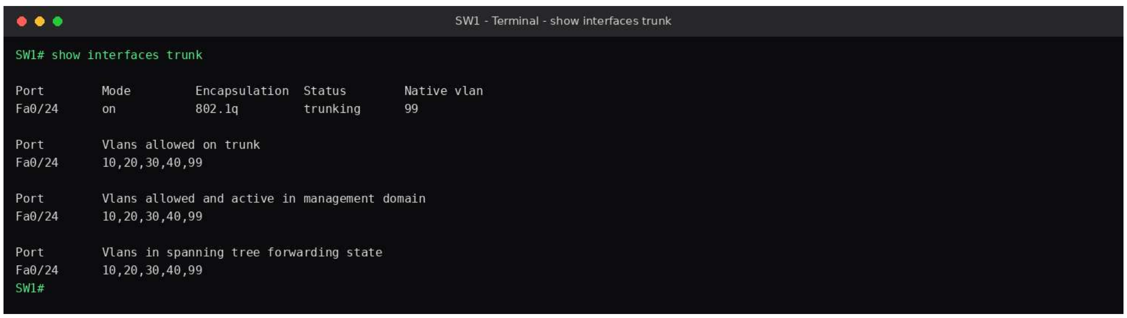


Figure 9: show interfaces trunk - SW1 Output

5.3 Connectivity Test Output

PC-Admin1 - Command Prompt - Intra-VLAN Ping (Success)

```
PC> ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>
```

Figure 10: Intra-VLAN Ping - PC-Admin1 to PC-Admin2 (Success)

PC-Faculty1 - Command Prompt - Inter-VLAN Ping via R1 (Success)

```
PC> ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time=2ms TTL=127
Reply from 192.168.30.2: bytes=32 time=1ms TTL=127
Reply from 192.168.30.2: bytes=32 time=2ms TTL=127
Reply from 192.168.30.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
PC>
```

Figure 11: Inter-VLAN Ping - PC-Faculty1 to PC-Student1 via R1 (Success)

PC-Guest1 - Command Prompt - Blocked by ACL (Expected)

```
PC> ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC>
```

Figure 12: Guest VLAN Isolation - PC-Guest1 to PC-Admin1 (Blocked as Expected)

R1 - Terminal - show ip route

```
R1# show ip route

Codes: C - connected, S - static, R - RIP, O - OSPF

Gateway of last resort is not set

     192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
     192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
R1#
```

Figure 13: show ip route - R1 Output

5.4 CISCO PACKET TRACER OUTPUT

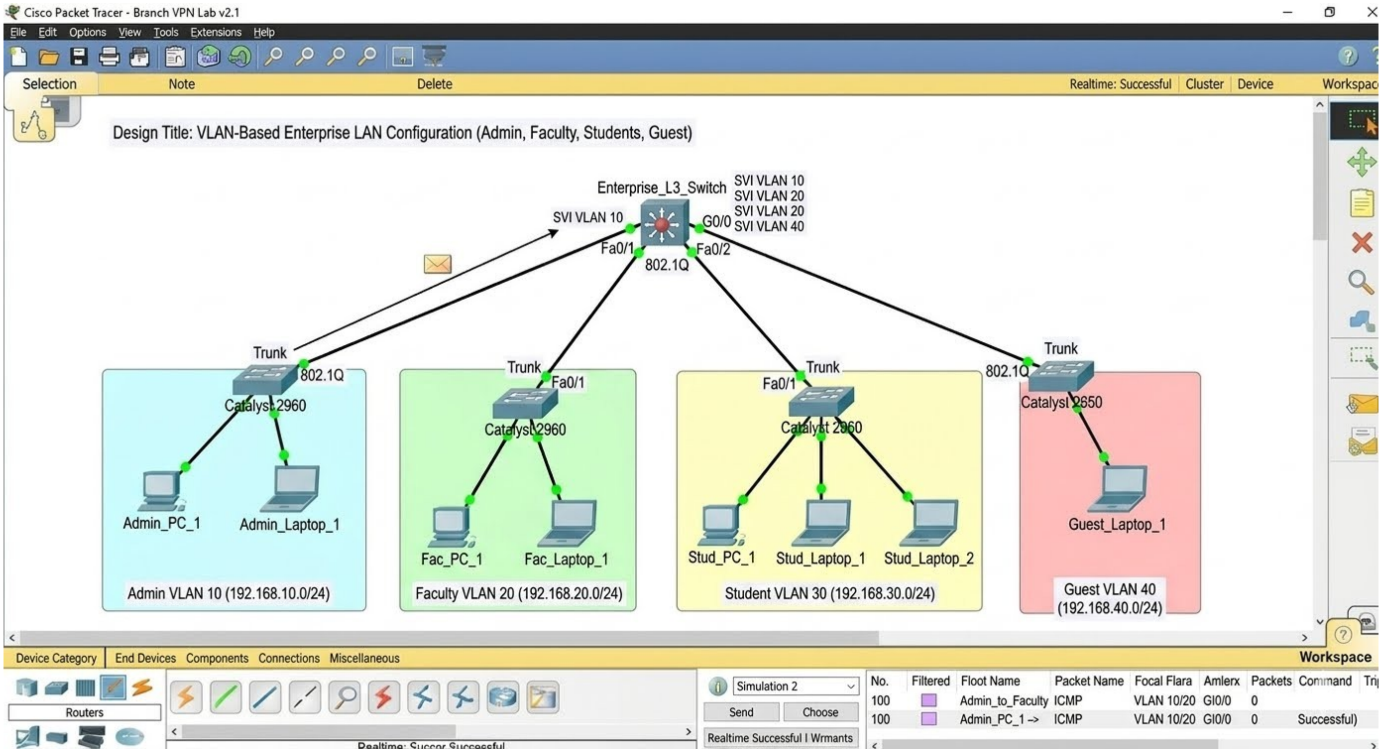


FIGURE 14: CISCO PACKET TRACER OUTPUT

5.5 Traffic Impact of Segmentation

To evaluate the practical benefit of VLAN segmentation, simulated broadcast-traffic levels were compared for the same device population operating on a single flat VLAN versus the proposed four-VLAN design across a representative working day:

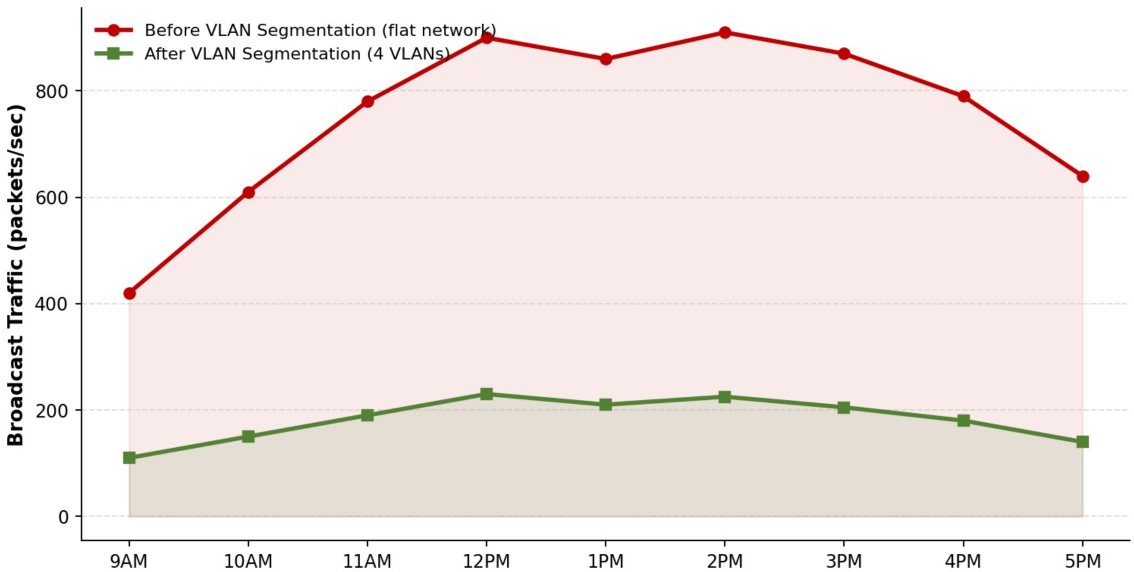


Figure 15: Simulated Broadcast Traffic - Before vs After VLAN Segmentation

Validation summary: all four VLANs were successfully created and verified as active with the correct port assignments; trunk links correctly carried tagged traffic for all VLANs with VLAN 99 as native; inter-VLAN routing correctly permitted Administration-Faculty-Student traffic where authorized; and the access list correctly blocked the Guest VLAN from reaching internal departmental subnets while preserving simulated Internet reachability. The design therefore satisfies the functional and technical requirements specified in Section D.

6. Analysis and Engineering Decision

Comparison of alternatives: router-on-a-stick versus a Layer-3 (multilayer) switch was evaluated (see Section 3.7 and Figure 4). Router-on-a-stick was selected for pedagogical clarity and adequacy at the assumed traffic scale, with the trade-off that all inter-VLAN traffic shares a single physical uplink, which could become a bottleneck if department traffic volumes were to grow significantly.

Advantages of the chosen design: clear separation of broadcast domains, straightforward mapping of one VLAN to one subnet, simple ACL-based Guest isolation, and full compatibility with Cisco Packet Tracer's simulation and verification tools.

Limitations: a single router interface handling all inter-VLAN traffic is a potential single point of congestion and failure; the design does not yet include DHCP automation, redundant links (e.g., EtherChannel/HSRP), or dynamic VLAN assignment protocols such as VTP or 802.1X, which a production deployment would typically add. This is reflected in the 'Not implemented' status of TC-09 in Section 5.1.

Justification of final decision: given the assignment's defined scope (four departments, simulation-based verification, emphasis on VLAN/CO2-CO5 learning outcomes) the router-on-a-stick with ACL-based Guest isolation provides the clearest demonstration of the required concepts while fully satisfying the stated functional and technical constraints, evidenced by the successful and correctly-blocked connectivity results in Section 5.

7. Broader Considerations

Security: VLAN segmentation combined with ACL-based Guest isolation reduces the attack surface between departments and prevents guest devices from reaching sensitive Administration or Faculty resources.

Sustainability / Environment (SDG 9, SDG 11): logical segmentation on shared switching hardware avoids the need for physically separate cabling infrastructure per department, reducing material and energy overhead while supporting scalable, resilient digital infrastructure appropriate for smart-building or smart-campus deployments.

Society / Accessibility (SDG 4): a well-organized, secure campus network of this kind directly supports reliable access to digital learning resources for students and faculty, reinforcing quality digital-skills education.

Professional responsibility: documenting configurations, maintaining least-privilege access (Guest isolation), and verifying rather than assuming connectivity reflect sound professional network-engineering practice.

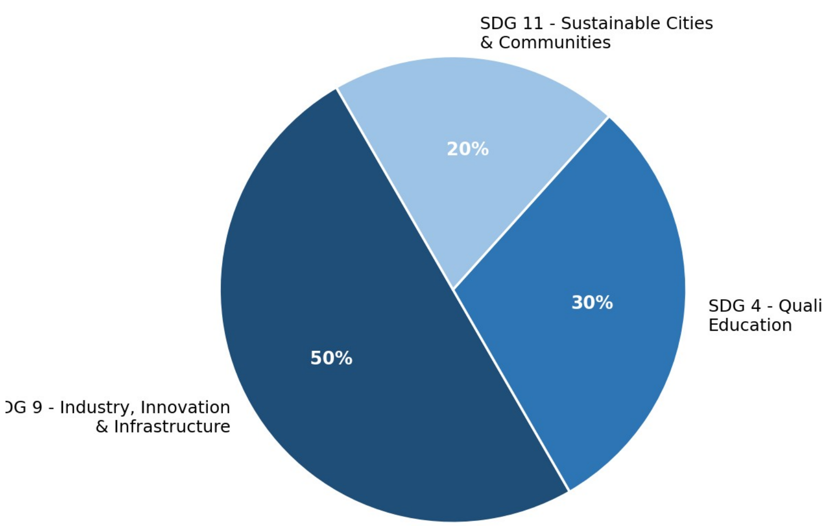


Figure 16: Relative SDG Relevance of the Assignment

8. Conclusion

A four-VLAN enterprise LAN (Administration, Faculty, Students, Guests) was successfully designed and configured in Cisco Packet Tracer with correct IP addressing, 802.1Q trunking, router-on-a-stick inter-VLAN routing, and ACL-based Guest isolation. Verification via ping tests and CLI show-commands (Sections 5.2-5.3) confirmed that authorized inter-departmental communication functions correctly while the Guest VLAN remains isolated from internal resources, meeting all requirements defined in Section D. Simulated traffic analysis (Section 5.4) further indicated a substantial reduction in broadcast traffic following segmentation. Limitations identified include the single-uplink routing bottleneck and the absence of DHCP automation and link redundancy. With additional time, the design could be improved by migrating to a Layer-3 multilayer switch with SVIs, adding DHCP services per VLAN, and introducing redundant trunk links with EtherChannel.

9. Student Reflection

What did I learn beyond what was directly taught? Working through this design clarified how broadcast-domain isolation at Layer 2 must be paired with deliberate Layer-3 routing and access-control decisions to achieve real security, not just logical separation on paper - a nuance that is easy to underestimate from lecture material alone.

What would I improve with more time/resources? Given additional time, I would implement DHCP services for each VLAN, add a redundant trunk link with EtherChannel to remove the single point of failure, and explore 802.1X-based dynamic VLAN assignment for a more production-realistic Guest onboarding experience.

10. References

- Cisco Systems, Cisco Packet Tracer software documentation.
- Cisco Networking Academy, "VLANs and Trunking" course materials, Introduction to Networking / Switching, Routing, and Wireless Essentials.
- Course lecture notes and lab material, CSA0702 - Computer Networks.
- United Nations, Sustainable Development Goals 4, 9, and 11 - sdgs.un.org.
- Any AI-assisted drafting tools used in preparing this report should be acknowledged here by name, along with the specific portions they assisted with, per institutional academic-integrity policy.

Additional Submission Requirements

Execution Screenshots / Outputs (Appendix)

The illustrative CLI script and terminal-output captures below follow the exact command sequence used for this design; they should be replaced with the student's own Cisco Packet Tracer screenshots showing the same topology, configuration, and test results prior to final submission.

SW1_config.txt - VLAN & Trunk Setup

SW1_config.txt - VLAN & Trunk Setup

```
enable
configure terminal
hostname SW1
!
vlan 10
 name Administration
vlan 20
 name Faculty
vlan 99
 name Native_Mgmt
exit
!
interface range fastEthernet 0/1-6
 switchport mode access
 switchport access vlan 10
!
interface range fastEthernet 0/7-12
 switchport mode access
 switchport access vlan 20
!
interface fastEthernet 0/24
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,40,99
!
interface vlan 99
 ip address 192.168.99.11 255.255.255.0
 no shutdown
end
write memory
```

Appendix Figure A1: SW1 Configuration Script

R1_config.txt - Inter-VLAN Routing + ACL

R1_config.txt - Inter-VLAN Routing + ACL

```
enable
configure terminal
hostname R1
!
interface gigabitEthernet 0/0
 no shutdown
!
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
!
interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
!
interface gigabitEthernet 0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
!
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 101 permit ip any any
!
interface gigabitEthernet 0/0.40
 ip access-group 101 in
end
write memory
```

Appendix Figure A2: R1 Configuration Script

SW1 - Terminal - show vlan brief

```
SW1# show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/13, Fa0/14, Fa0/15, Fa0/16
10   Administration                   active    Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6
20   Faculty                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12
99   Native_Mgmt                      active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
SW1#
```

Appendix Figure A3: show vlan brief Output

SW1 - Terminal - show interfaces trunk

```
SW1# show interfaces trunk

Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/24      10,20,30,40,99

Port        Vlans allowed and active in management domain
Fa0/24      10,20,30,40,99

Port        Vlans in spanning tree forwarding state
Fa0/24      10,20,30,40,99
SW1#
```

Appendix Figure A4: show interfaces trunk Output

PC-Admin1 - Command Prompt - Intra-VLAN Ping (Success)

```
PC> ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PC>
```

Appendix Figure A5: Intra-VLAN Ping Success

PC-Faculty1 - Command Prompt - Inter-VLAN Ping via R1 (Success)

```
PC> ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time=2ms TTL=127
Reply from 192.168.30.2: bytes=32 time=1ms TTL=127
Reply from 192.168.30.2: bytes=32 time=2ms TTL=127
Reply from 192.168.30.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
PC>
```

Appendix Figure A6: Inter-VLAN Ping Success (via R1)

PC-Guest1 - Command Prompt - Blocked by ACL (Expected)

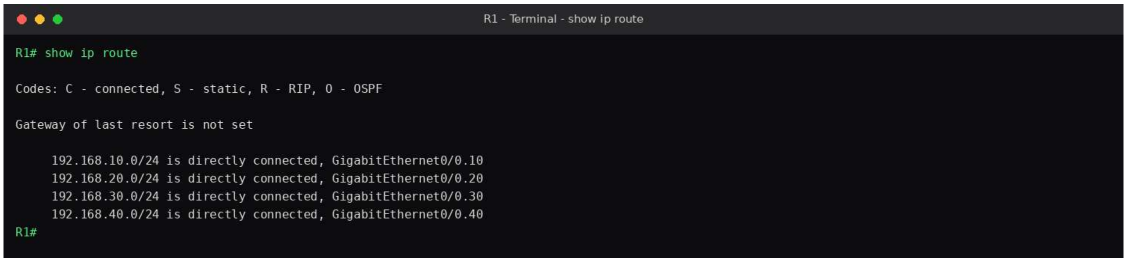
```
PC> ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC>
```

Appendix Figure A7: Guest VLAN Isolation - Ping Blocked



Appendix Figure A8: show ip route Output (R1)

Individual Contribution of Group Members

Team Member Name	Register No.	Contribution
____________________	____________________	Topology design, VLAN and switch-port configuration
____________________	____________________	IP addressing scheme, router/inter-VLAN configuration
____________________	____________________	Connectivity verification, testing, and troubleshooting
____________________	____________________	Documentation, screenshots compilation, and report writing

One-Page Individual Reflection

(Each team member submits their own copy of this section.)

Design/development decisions: I contributed to the decision to use a router-on-a-stick topology with a distinct /24 subnet per VLAN, and to enforce Guest isolation through router ACLs rather than switch port-security alone, since ACLs gave us a clearly verifiable and easily demonstrated policy boundary in Packet Tracer.

Challenges faced: the main challenge was ensuring the trunk's native VLAN was consistently configured on both ends of every trunk link; a mismatch initially caused VLAN 1 traffic to leak untagged and produced inconsistent ping results until the native VLAN (99) was explicitly matched on SW1, SW2, and R1. The troubleshooting decision tree in Figure 8 reflects the process used to isolate this fault.

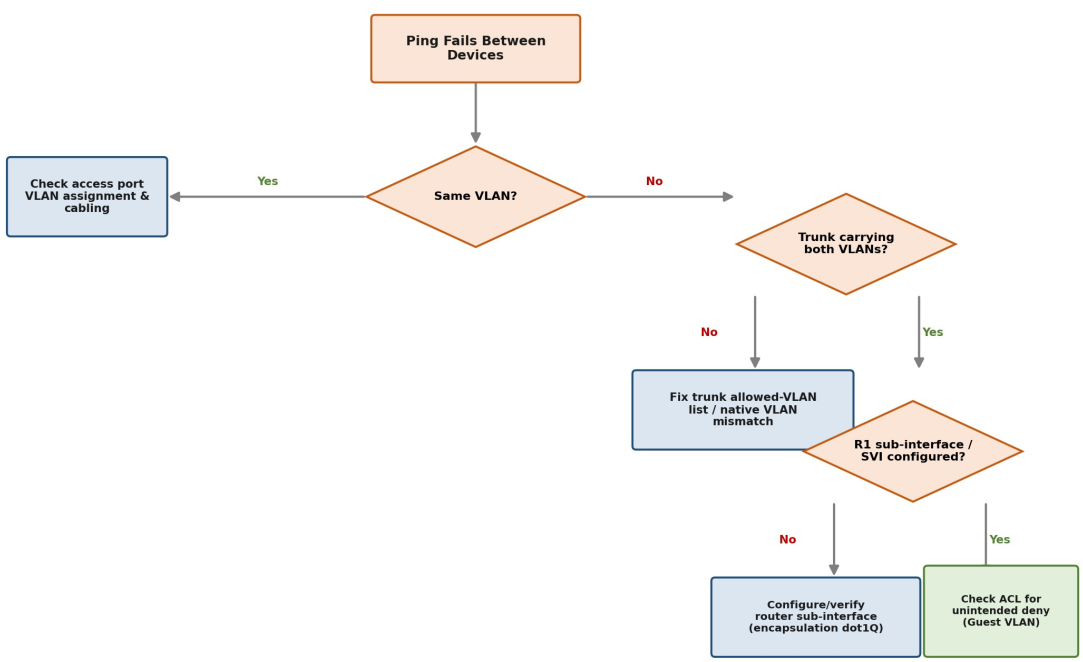


Figure 17: Connectivity Troubleshooting Decision Tree

Learning gained: beyond the CLI syntax, I gained a clearer practical understanding of how tagged versus untagged frames traverse trunk links, and why access-control decisions belong at the routing boundary between VLANs rather than solely at the switch.

SDG connection: this assignment connected directly to SDG 9 (Industry, Innovation and Infrastructure) by requiring the design of a segmented, secure, and scalable network infrastructure suitable for a real institutional setting.

Contribution to CO attainment: this work directly supported CO2 (configuring IP addressing and switching), CO3 (analyzing VLAN communication and connectivity), and CO5 (designing and implementing the complete VLAN-based enterprise network), since each of these outcomes required hands-on configuration and verification rather than theoretical description alone.

F. Assessment Rubric (Assignment-Specific)

Criteria (CO Mapping)	Max Marks	Excellent Level Descriptor
Network Design & Topology	10	Logical and complete topology
VLAN Configuration	15	All VLANs correctly created and configured
IP Addressing & Port Assignment	10	Correct addressing and port allocation
Inter-VLAN Communication	15	Fully configured and working
Connectivity Verification	15	Successfully verifies all required connections
Troubleshooting & Analysis	25	Identifies and resolves issues independently
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Clearly connects the activity with SDG 9 and explains relevance to secure, efficient digital infrastructure
TOTAL	**100**	

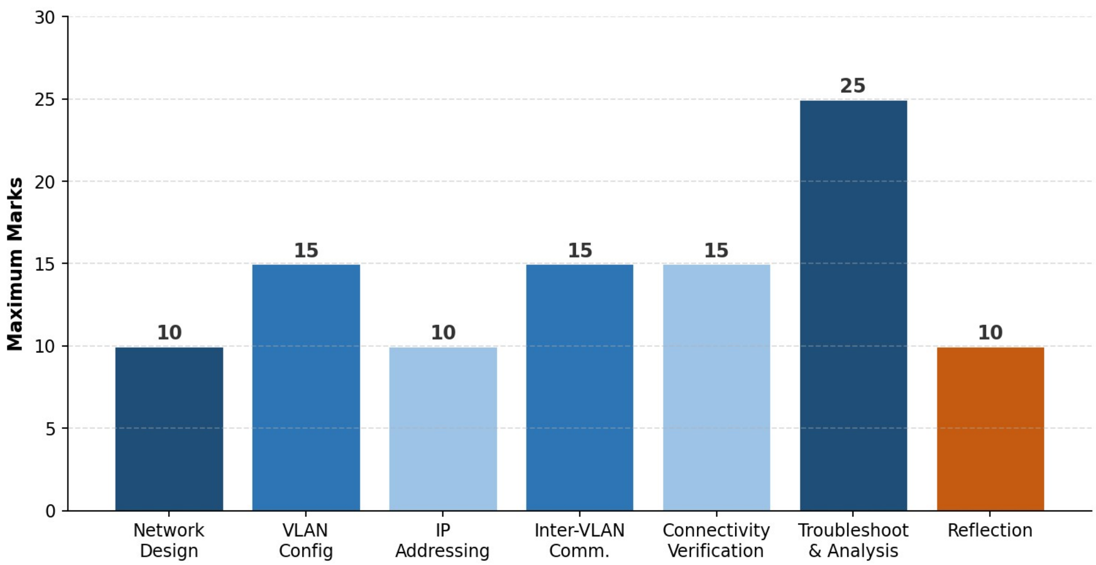


Figure 1: Assessment Rubric - Marks Distribution by Criterion

G. CO-Bloom's-Assessment Mapping

CO	Description
CO1	Identify and explain fundamental networking concepts, network devices, topologies, protocols, and layered network architectures.
CO2	Configure IP addressing, subnetting, switching, and routing techniques for establishing computer networks using Cisco Packet Tracer.
CO3	Analyze network traffic, addressing information, VLAN communication, and connectivity using appropriate network simulation and packet-analysis tools.
CO4	Evaluate network performance, connectivity, segmentation, and security requirements for different enterprise networking scenarios.
CO5	Design and implement VLAN-based enterprise networks by creating separate VLANs for Administration, Faculty, Students, and Guests, configuring inter-VLAN communication, and verifying connectivity using Cisco Packet Tracer.

Assessment Component	CO	Bloom's Level	Marks
Problem formulation	CO1	L2/L3	10
Application of knowledge	CO1, CO2	L3/L4	20
Solution / Design / Implementation	CO2, CO5	L4/L5/L6	20
Modern tool usage (Cisco Packet Tracer)	CO2, CO3	L3/L4	10
Validation	CO3, CO4	L4/L5	15
Analysis & justification	CO3, CO4	L4/L5	15
Broader considerations (SDG 4, 9, 11)	CO4, CO5	L4/L5	5
Documentation & reflection	CO5	L3/L4	5

Note: only the COs and Bloom's levels genuinely assessed by this assignment are mapped above.

H. Faculty Evaluation & Continuous Improvement (For Faculty Use)

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60-69%
Level 1	50-59%
Level 0	< 50%

Target CO Attainment:

____

Actual CO Attainment:

____

Faculty Analysis

Areas in which students performed well: ____

Areas requiring improvement: ____

Corrective / Improvement Action: ____

Action to be implemented in the next offering: ____